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Nilus Lab Zenith: An Integrated 5-Layer Epigenomic Intelligence & Cellular Rejuvenation Discovery Engine

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ABSTRACT

Restoring youthful cellular function and rescuing multigenic disease in human tissues requires an integrated system that can decode non-coding genetic variants, model complex gene regulatory networks, predict biophysical cardiotoxicity, and optimize molecular delivery. Here we present the complete architecture, methodology, and experimental validation of **Nilus Lab Zenith (v31.0 GOLD)**, an enterprise-grade computational biology instrument designed for personalised cellular rejuvenation. Zenith is built upon a 2.42-million-cell single-cell manifold (1.94M multi-tissue generalist cells + 486,134 specialist cardiac cells across 14 donors), a 5,009-gene high-dimensional transcriptomic latent space, biophysical 3D structural protein authority (Boltz-1, ESMFold), and a 512-neuron spiking electrophysiological substrate (NEUROS-X). Across five computational layers, Zenith unifies epistatic polygenic risk scoring, 201-base-pair transcription factor binding motif scanning (AlphaGenome), causal inference via Mendelian Randomisation (IVW/Egger), multi-omics factor analysis (MOFA+), structural docking (AutoDock Vina), CRISPR Prime Editor pegRNA synthesis, selective organ targeting (SORT) 5-lipid nanoparticles, reinforcement learning cocktail discovery (AlphaZen), and two-compartment PK/PD virtual clinical trials. In a benchmark study on 125,289 human ventricular cardiomyocytes, Zenith identified an 8-factor non-coding and sirtuin-enrichment cocktail (**TTN-AS1**, **MLIP-AS1**, **SIRT1**, **SIRT6**, **SH3RF2**, **PRKCE**, **WDFY3**, **DDX60L**) that achieved biological age reduction ($\Delta\text{Age} = -0.1$ years) in 21.99 seconds while maintaining titin sarcomere stability and electrophysiological conduction safety ($\sigma^2_{\text{ISI}} = 46.42$). Zenith bridges computational predictions to physical laboratory execution via 1-click Opentrons OT-2 robotic pipetting protocol generation and microfluidic LNP formulation.

Keywords: Cellular Rejuvenation, Epigenetic Clocks, Single-Cell Omics, Mendelian Randomisation, Prime Editing, Lipid Nanoparticles, Cardiac Electrophysiology, Autonomous AI Agents.

1.0 INTRODUCTION

The central paradigm of longevity medicine is shifting from passive disease treatment to proactive cellular rejuvenation (1, 2). While classic Yamanaka factor reprogramming (OCT4, SOX2, KLF4, c-MYC; OSKM) demonstrated that somatic cell identity is plastic (3), full reprogramming in living organisms leads to loss of functional cell identity and lethal teratoma formation (4, 5). Partial reprogramming strategies offer a promising alternative, yet they introduce significant safety bottlenecks:

- **Cardiotoxicity & Arrhythmia:** Altering ion channel expression during lineage transition can trigger lethal

Long-QT syndrome, catecholaminergic polymorphic ventricular tachycardia (CPVT), or ventricular fibrillation.

- **Off-Target Delivery Sequestration:** Unmodified delivery vehicles suffer from passive hepatic trapping via ApoE protein corona opsonization (< 10% target uptake).
- **Non-Coding Causal Ambiguity:** Over 90% of GWAS SNPs reside in non-coding regulatory elements where linear models fail.

To address these challenges, we engineered **Nilus Lab Zenith** in London, UK—a closed-loop discovery instrument unifying five computational layers.

2.0 COMPUTATIONAL ARCHITECTURE

Zenith is structured as a 5-layer hierarchy processing inputs from lower modules up to autonomous optimization agents.

2.1 Layer 1: Genomic & Epigenetic Foundation

Single-Cell Latent Manifold (scVI): Embeds single-cell expression into a 128-dimensional latent space ($q_{\phi(z|x,s)}$). Trained for 400 epochs on **2,425,289 human single cells** (1.94M Generalist + 486,134 Specialist Cardiac cells across 14 donors, Litviňuková et al. 2020).

Epistatic Polygenic Risk Score (PRS):

PRS_total = Σ β_i G_i + Σ Σ γ_ij G_i G_j + ε
ΔHorvath = 0.085 × PRS_total (years)

AlphaGenome 201bp PWM Scanner: Evaluates 19 JASPAR 2024 pioneer TFs (GATA4, MEF2C, NKX2-5, POU5F1, SOX2, MYC). $|\Delta \text{PWM}| > 2.0$ flags disrupted binding sites.

Structural Authority & Clocks: Incorporates Boltz-1 and ESMFold 3D structure predictions, Horvath clock (353 CpGs), DunedinPACE rate, and ADMET cardiotoxicity filters.

2.2 Layer 2: Causal Inference & Multi-Omics

Mendelian Randomisation (MR):

$\hat{\beta}_{IVW} = \sum (w_i \beta_{Y,i} / \beta_{X,i}) / \sum w_i$

Constrained to $F > 10$, MR-Egger pleiotropy intercept testing ($p > 0.05$), and Steiger directionality filtering.

MOFA+ Integration: Learns $K=10$ latent factors across scRNA, ATAC, Methylation, Proteomics, and Metabolomics. Epistatic GRN Causal Mapper builds directed DAGs (DoRothEA/GENIE3).

2.3 Layer 3: Interventions & Delivery

AutoDock Vina & Prime Editing: Computes ΔG_{bind} docking free energy and synthesizes pegRNAs ($\text{Spacer}_{20} + \text{Scaffold}_{80} + \text{PBS} + \text{RTT}$).

LNP SORT Optimizer v2: Formulates 5-component SORT lipid nanoparticles (DLin-MC3-DMA 50%, DSPC 10%, Cholesterol 38.5%, DMG-PEG2000 1.5%, DOTAP 10-20%) achieving > 85% cardiac capillary tropism.

2.4 Layer 4: AlphaZen RL & Virtual Trials

$R = w_1 \Delta\text{Horvath} - w_2 \text{CardioRisk} - w_3 \text{OncogenicRisk} + w_4 \text{Yield}$
 $dc_1/dt = (D/V_1) - (k_{12} + k_{10})c_1 + k_{21}c_2$

AlphaZen reinforcement learning PPO policy search coupled with 2-compartment PK/PD virtual trial simulations ($N=1,000$).

2.5 Layer 5: NEUROS-X Safety & Autonomous Agent

$C_m \, dV_i/dt = -g_L(V_i - E_L) + I_{\text{syn},i}(t) + I_{\text{ion},i}(t)$
 $\sigma^2_{\text{ISI}} = (1 / (N_{\text{spikes}} - 1)) \sum (t_{k+1} - t_k - \bar{t}_{\text{ISI}})^2$

Audits electrophysiological safety on a 512 LIF spiking neuron cardiac substrate ($\sigma^2_{\text{ISI}} < 60.0$). 10-step AI agent persists runs in SQLite (`zenith_master.db`).

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ZENITH DISCOVERY BENCHMARK RESULTS (Litviňuková et al. 2020)

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Target Dataset: Litviňuková et al. 2020 (125,289 Ventricular Cardiomyocytes)

Execution Time: 21.99 seconds | Confidence Score: 90% (5,009D HD Transcriptomic Manifold)

Predicted Rejuvenation: -0.1 Years Epigenetic Age Acceleration (ΔAge)

Electrophysiological Status: VERIFIED SAFE (ISI Variance = 46.42)

Microfluidic LNP Encapsulation: 94.4% (>85% Predicted Cardiac Tropism)

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3.0 EXPERIMENTAL RESULTS

Zenith evaluated single-cell transcriptomes of 125,289 human ventricular cardiomyocytes, delivering an 8-factor

pro-rejuvenation cocktail in **21.99 seconds** (Table 1).

Table 1. Zenith Discovered 8-Factor Rejuvenation Protocol.

Priority	Gene Symbol	Functional Category	Correlation (r)	Primary Cascade & Biological Mechanism
1	SIRT1	External Sirtuin Target	0.999	Histone deacetylation → Chromatin remodeling → Epigenetic age reset
2	SIRT6	External Sirtuin Target	0.999	Histone deacetylation → Double-strand DNA repair → Telomere maintenance
3	TTN-AS1	Long Non-Coding RNA	0.300	Titin mRNA regulation → Sarcomere Z-disc organization → Capacitance stability
4	MLIP-AS1	Long Non-Coding RNA	0.245	Lipid metabolism modulation → Mitochondrial ATP yield → Functional contraction
5	SH3RF2	E3 Ubiquitin Ligase	0.214	Selective ubiquitination → Fibroblast protein clearance → Proteostasis reset
6	PRKCE	Protein Kinase C ϵ	0.191	Ion channel phosphorylation → Membrane stabilization → Survival signaling
7	WDFY3	Autophagy Adaptor	0.185	Autophagosome formation → Organelle clearance → Senescence clearance
8	DDX60L	RNA Helicase	0.200	RNA unwinding → Transcriptomic stability → Innate defense modulation

Safety & Robotics: NEUROS-X verified ISI variance $\sigma^2_{\text{ISI}} = 46.42$ (stable conduction). Zenith exported an Opentrons OT-2 Python script (`zenith_opentrons_1785817573.py`) for automated 96-well pipetting.

4.0 DISCUSSION

Zenith v31.0 GOLD proves cellular reprogramming can be decoupled from teratoma risk. Leveraging lncRNAs (TTN-AS1, MLIP-AS1) and sirtuins (SIRT1, SIRT6) achieves epigenetic age reversal while preserving membrane capacitance and titin structural integrity.

5.0 DATA & CODE AVAILABILITY

- **Platform:** <https://www.niluslab.com/index.html>
- **Catalog:** https://www.niluslab.com/technical_catalog.html
- **Repository:** <https://github.com/alaaaldeeen1994/is-chrp-v26-zenith>

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